Postsecondary Education in Tennessee: An ROI Analysis

By Dennis Vaughan

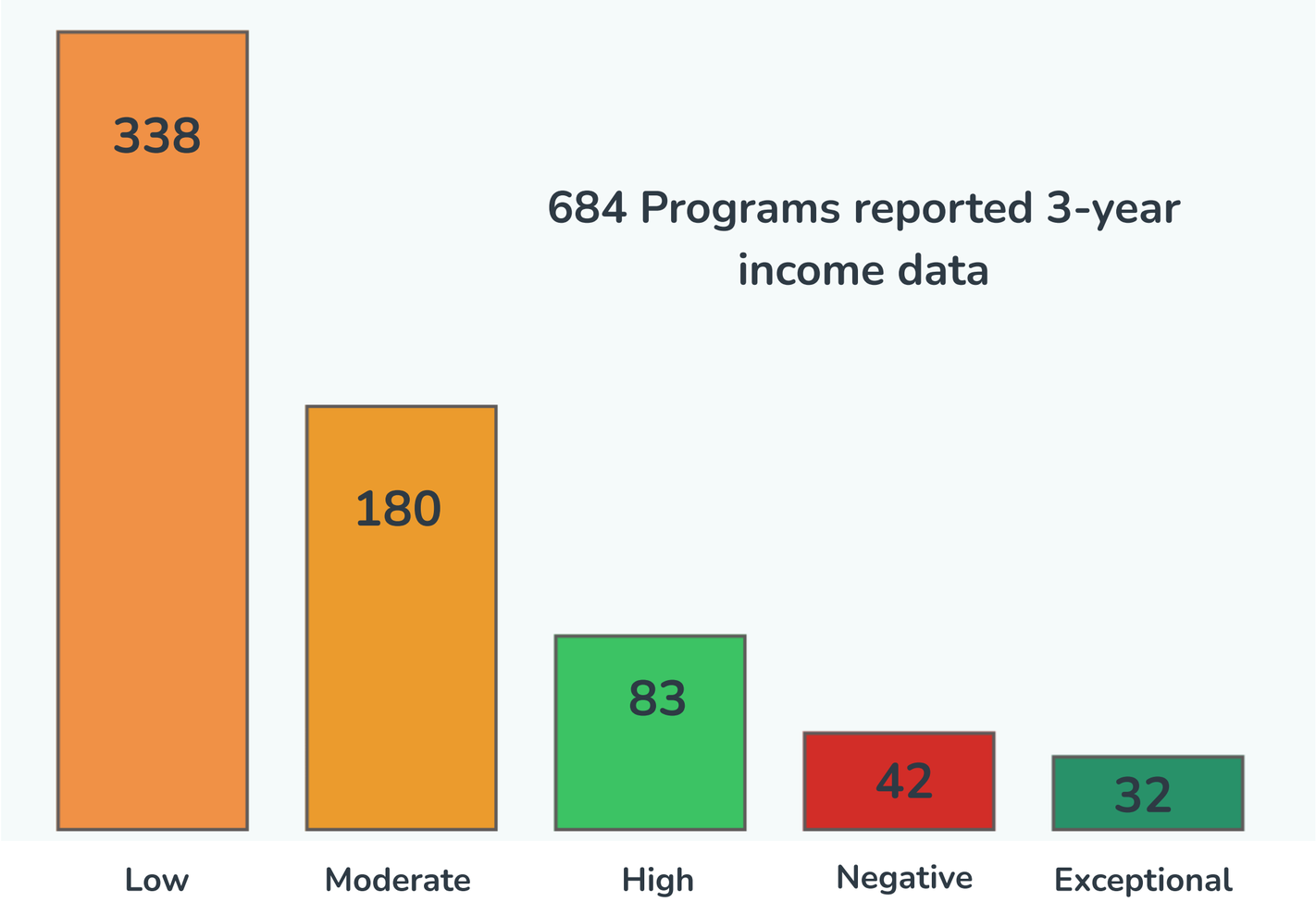
SUMMARY

This project seeks to analyze the return on investment (ROI) of postsecondary programs in the state of Tennessee using College Scorecard and IPEDS completions data. ROI is based on reported earnings three years after program completion. The goal is to explore how early career earnings differ by degree type and program of study.

DEFINITIONS

Return on Investment (ROI):	Measure of (median earnings - median cost) / median cost
ROI Score Categories:	Programs were categorized based on calculated ROI values and grouped into tiers.
High / Exceptional Share:	Percentage of programs within a CIP Family classified as High or Exceptional ROI
Negative Share:	Percentage of programs classified as Negative ROI
CIP Code:	Classification of Instructional Program
CIP Family:	CIP Code group name

PROGRAM COUNT BY ROI CATEGORY



TOP HIGH-ROI CIP FAMILIES

CIP Family	HIgh/Exceptional Share	Program Count	Areas of Study and Employment
Engineering Technologies & Technicians	50%	16	Architectural, Civil, Mechanical, Nuclear, Computer, & Construction
Engineering	41%	27	Architectural, Civil, Mechanical, Nuclear, Computer, & Construction
Health Professionals & Clinical Sciences	32%	158	Chiropractic, Dental, Rehab, Pharmacy, Veterinary, Medical & Pre-Med
Computer & Information Science	27%	22	Programming, Processing, System Analysis, Networking, & Support
Liberal Arts	27%	22	Liberal Arts & Sciences, General Studies, Humanities
Precision Production	27%	15	Leatherworking, Upholstery, Woodworking, Precision Metal, Boilermaking

BOTTOM HIGH-ROI CIP FAMILIES

CIP Family	HIgh/Exceptional Share	Program Count	Areas of Study and Employment
Security & Protective Services	10%	21	Criminal Justice, Corrections, Fire Protection, Homeland Security
Business Management & Marketing	11%	94	Accounting, Business Admin, HR, Marketing, Real Estate, & Hospitality
Mechanic & Repair Technician	26%	43	Mechanical, Electrical, & Vehicle Repair, HVAC, Heavy Equipment
Computer & Information Science	27%	22	Programming, Processing, System Analysis, Networking, & Support
Liberal Arts	27%	22	Liberal Arts & Sciences, General Studies, Humanities
Precision Production	27%	15	Leatherworking, Upholstery, Woodworking, Precision Metal, Boilermaking

ROI DISPERSION ACROSS CIP FAMILIES

Health Professionals & Clinical Sciences

- High / Exceptional Share: 32%
- Negative Share: 13%
- Program Count: 158
- Grade: Volatile

Business Management & Marketing

- High / Exceptional Share: 11%
- Negative Share: 2%
- Program Count: 94
- Grade: Stable Low

Engineering Technologies & Technicians

- High / Exceptional Share: 50%
- Negative Share: 6%
- Program Count: 16
- Grade: *ELITE*

Security & Protective Services

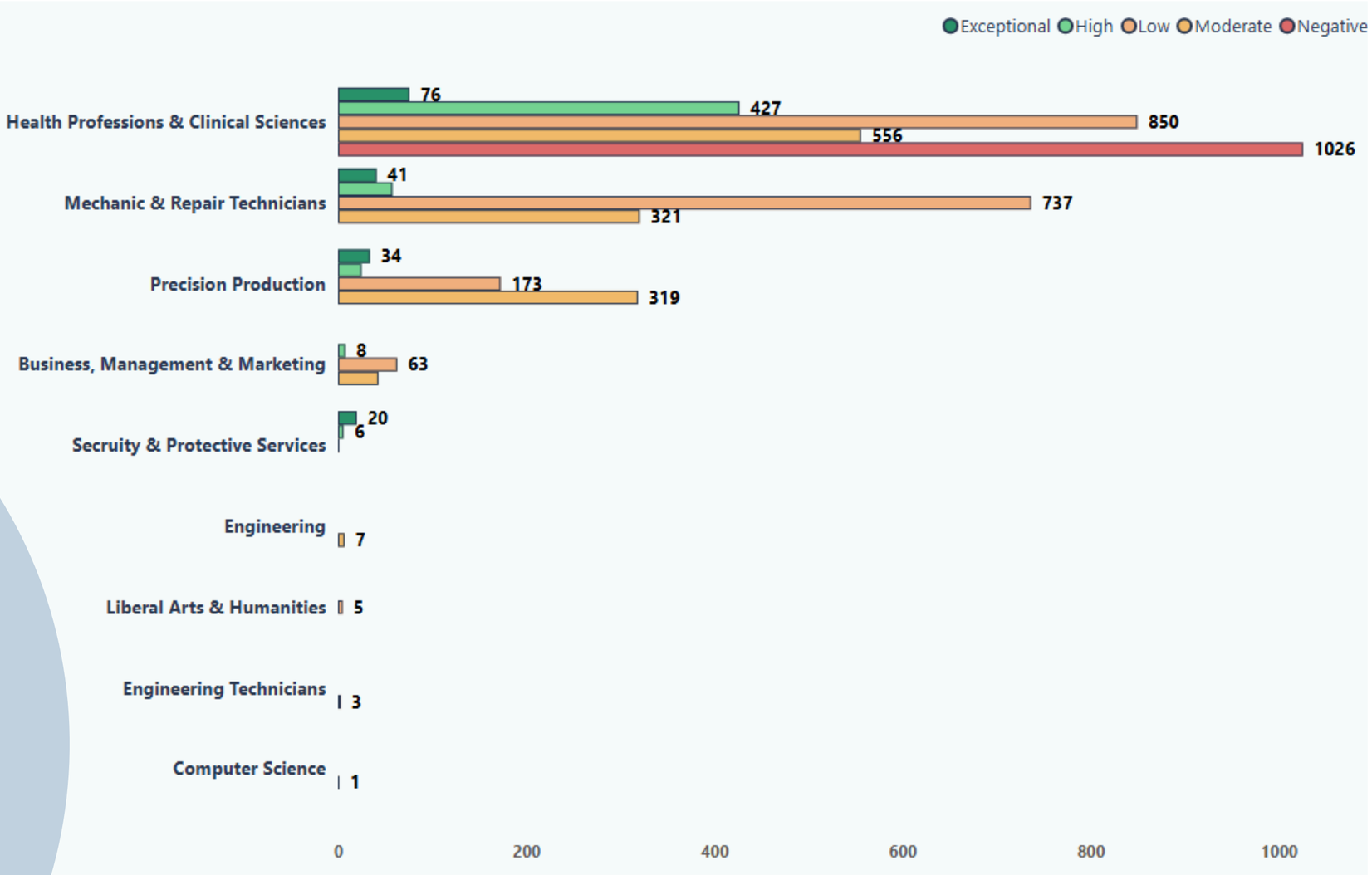
- High / Exceptional Share: 10%
- Negative Share: 6%
- Program Count: 21
- Grade: Volatile

CREDENTIAL DISTRIBUITON BY CIP FAMILY

CIP Family	Associate	Bachelor	Certificate	Total
Health Professions & Clinical Sciences	58	40	60	158
Business, Management & Marketing	18	64	12	94
Mechanic & Repair Technicians	2		41	43
Engineering		27		27
Computer Science	7	14	1	22
Liberal Arts & Humanities	14	8		22
Secruity & Protective Services	5	14	2	21
Engineering Technicians	10	6		16
Precision Production			15	15

High and low ROI programs appear at every credential level across the featured fields, highlighting variation within each CIP Family.

PRODUCTION FROM FEATURED PROGRAMS



CONCLUSION

Ultimately, the results suggest that educational outcomes vary widely across programs, and better aligning enrollment with high-return programs could improve economic outcomes for students.

Where students enroll matters just as much as what they study.

Thank you!

For follow up or additional analysis, here is how to reach me:

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